TWIN CITIES LEDGER

METRO MINNEAPOLIS AND ST. PAUL

WINTER ROADS

Brine before the storm

Plow crews are spraying salt water on the roads before the snow falls. It uses less salt, and it is catching on.

By Owen Brandt, Staff Writer
January 2018

Hours before the first flakes, the trucks go out with tanks instead of salt hoppers. They lay thin stripes of brine on bridges, ramps and hills, and when the snow arrives it has a harder time bonding to the pavement.

Road crews call it anti-icing, and it turns the usual order of winter work around. For decades the routine was to let the snow fall, plow it and spread rock salt on whatever packed down into ice. More crews now put the salt down first, already dissolved, so it is working the moment the storm starts.

Crew supervisors say the payoff is simple: salt that stays on the road instead of bouncing into the ditch. “Every ton we don’t throw is a ton we don’t have to buy,” one county maintenance supervisor said, “and a ton that doesn’t end up in a lake.”

That last part is why people who never think about plow routes have started paying attention. Salt does not stay on the road. Snowmelt carries it into storm drains, and storm drains empty into lakes, ponds, wetlands and streams. In the water the salt splits apart, and the chloride stays. It does not settle to the bottom, and it does not break down.

A teaspoon of road salt is enough to permanently pollute 5 gallons of water, according to the Minnesota Pollution Control Agency. The state’s chronic standard for chloride, the level above which long exposure harms fish, insects and other aquatic life, is 230 milligrams per liter. Once chloride is in a lake, the agency says, there is no practical way to take it back out.

The people who look after the water have started watching the people who look after the roads. The Metro Lakes Watershed District, which monitors lakes, ponds and streams in its member cities, says it plans to begin testing five of its lakes for chloride next year.

Brine itself is cheap to make. At a county shop in the east metro, a machine about the size of a garden shed churns rock salt and water together and pumps the result into storage tanks. The operator checks each batch with a hydrometer and aims for 23.3 percent salt by weight, the mix that stays liquid at the lowest temperature: about 6 degrees below zero.

From there, a tank truck sprays it in stripes at about 40 gallons per lane mile. A gallon of brine at that strength holds a little more than 2 pounds of salt, so a full pass puts down about 90 pounds a mile. Spreading dry rock salt on a road that has already iced over can take several times that.

Bridges and ramps get the first stripes because they freeze first. With cold air above and below the deck, a bridge can ice over while the road leading up to it is still wet.

Dry salt also has to pull water out of the snow before it melts anything, and until it does, it bounces. On a cold, dry road, the spinner on the back of the truck and the traffic behind it throw a share of it onto the boulevard.

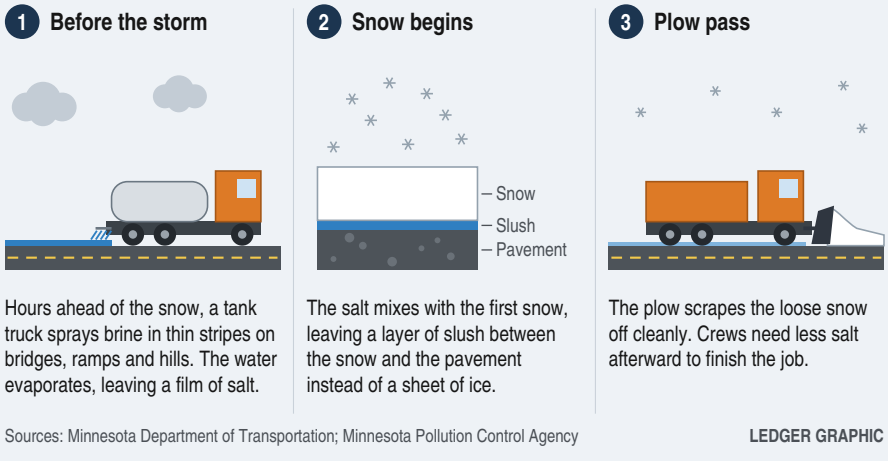
“You can watch it go,” said a plow driver who has run the same route for 14 winters and asked not to be named because the department had not authorized the interview. “You’re salting the grass.”

Brine does not bounce. When the water

See BRINE on page 2

HOW ANTI-ICING WORKS

Crews spray salt brine on the pavement before the snow starts, so ice has a harder time bonding to the road.



Sources: Minnesota Department of Transportation; Minnesota Pollution Control Agency LEDGER GRAPHIC

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BRINE Continued from page 1

Less salt, but still salt

evaporates, it leaves a film of salt on the pavement, the white stripes drivers notice on dry days before a storm. Supervisors say some drivers call in to report a truck that is leaking.

Many crews now also wet their rock salt with brine as it leaves the truck, which helps it stick to the road and start melting sooner. The Minnesota Department of Transportation has used brine on state highways for years, and a growing number of cities and counties have followed.

Not every agency is sold. Brine takes new equipment and training, and it does not work in every kind of storm. But the crews that have tried it say they would not go back.

If a storm starts as rain, the rain washes the brine away. If the forecast calls for blowing and drifting snow, crews hold off, because a wet road can catch snow that would otherwise blow across it. And when the pavement is colder than about 15 degrees, plain salt works slowly whether it goes down wet or dry, so crews turn to sand or to blends with other chemicals.

Timing matters, too. Brine sprayed too many days ahead wears off under traffic, and brine sprayed onto wet pavement runs off. “The forecast has to be right, and then we have to be right about the forecast,” the county supervisor said.

So the call is usually made the day before. Supervisors watch the forecasts and the state’s roadside weather stations, which report pavement temperatures on major highways, and many trucks now carry infrared thermometers that read the road surface as they drive.

Then there is the equipment. Tanks, pumps and spray bars have to fit on trucks that spend the rest of the winter plowing, and someone has to run the brine maker. For a small city with a handful of trucks, that is a real cost, and the savings take a few winters to show up.

Brine is also still salt. It cuts the amount that goes down; it does not eliminate it.

> *“Every ton we don’t throw is a ton we don’t have to buy.”*
>
> COUNTY MAINTENANCE SUPERVISOR

Every pound sprayed on a bridge deck still ends up somewhere, and the corrosion that eats at car bodies and bridge steel works the same whether the salt arrived wet or dry.

Nobody who works with brine claims it solves the chloride problem. City and county trucks are only part of the picture. Parking lots, sidewalks and private roads are salted by property managers and contractors, and much of that salt still goes down dry. The pollution control agency offers voluntary training for public crews and private applicators alike on using less salt without giving up safety.

On a gray afternoon last week, the plow driver filled the truck’s tank at the shop and went out to stripe the bridges on the route, hours ahead of a storm that was not due until after dark. By the time the first flakes fell, that part of the job was done.

“You don’t win a storm at 3 in the morning,” the driver said. “You win it the afternoon before.” ■

Owen Brandt covers science and the environment for the Ledger.

BY THE NUMBERS

23.3	Percent salt in the brine most crews mix. At that strength it stays liquid down to about 6 degrees below zero.
2.3	Pounds of salt in a gallon of that brine.
40	Gallons per lane mile the county crews in this story spray before a storm, about 90 pounds of salt.
230	Milligrams of chloride per liter: the state’s chronic standard for lakes and streams.
1	Teaspoon of road salt is enough to permanently pollute 5 gallons of water, the Minnesota Pollution Control Agency says.

Sources: Minnesota Pollution Control Agency; Minnesota Department of Transportation; Ledger reporting

SALTING AT HOME

Homeowners can cut back, too. Tips from the Minnesota Pollution Control Agency:

Shovel first.
The more snow you clear by hand, the less salt it takes, and the better that salt works.

A mug is plenty.
A heaping 12-ounce coffee mug holds about a pound of salt, enough for a 20-foot driveway or about 10 sidewalk squares.

Space it out.
Leave about 3 inches between grains. Using more will not melt the ice any faster.

Know the limit.
Below about 15 degrees, salt barely works. Use sand for traction instead.

Sweep up.
Salt left on dry pavement is not doing any work. Sweep it up and use it again.

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